

Homework 8

Problem 1. Let $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$.

- (a) We have $\|f\|_2 = 0 \iff f = 0$.
- (b) We have $|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2$.
- (c) We have $\|f + g\|_2 \leq \|f\|_2 + \|g\|_2$.
- (d) If $\langle f, g \rangle = 0$, then $\|f + g\|_2^2 = \|f\|_2^2 + \|g\|_2^2$.
- (e) We have $\|cf\|_2 = |c| \|f\|_2$ for all $c \in \mathbb{C}$.

Proof of (a). By Lemma 5.2.5 we have $\|f\|_2 = 0 \iff \sqrt{\langle f, f \rangle} = 0 \iff \langle f, f \rangle = 0 \iff f = 0$. □

Proof of (b). We have two cases.

Case 1: $\|g\|_2 = 0$. Then we have $|\langle f, g \rangle| = |\langle f, 0 \rangle| = 0 = \|f\|_2 \|0\|_2 = \|f\|_2 \|g\|_2$.

Case 2: $\|g\|_2 \neq 0$. Notice that

$$\begin{aligned}
 \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, g \right\rangle &= \langle f, g \rangle - \frac{\langle f, g \rangle}{\|g\|_2^2} \langle g, g \rangle \\
 &= \langle f, g \rangle - \langle f, g \rangle \frac{\|g\|_2^2}{\|g\|_2^2} \\
 &= \langle f, g \rangle - \langle f, g \rangle \\
 &= 0.
 \end{aligned} \tag{1}$$

Further, by equation (1) we have

$$\begin{aligned}
 \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, \frac{\langle f, g \rangle}{\|g\|_2^2} g \right\rangle &= \overline{\left(\frac{\langle f, g \rangle}{\|g\|_2^2} \right)} \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, g \right\rangle \\
 &= 0.
 \end{aligned} \tag{2}$$

Then,

$$\begin{aligned}
 0 &\leq \left\| f - \frac{\langle f, g \rangle}{\|g\|_2^2} g \right\| \\
 &= \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, f - \frac{\langle f, g \rangle}{\|g\|_2^2} g \right\rangle \\
 &= \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, f \right\rangle - \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, \frac{\langle f, g \rangle}{\|g\|_2^2} g \right\rangle \\
 &\stackrel{(2)}{=} \left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2} g, f \right\rangle.
 \end{aligned} \tag{3}$$

Then we have,

$$\begin{aligned}
\left\langle f - \frac{\langle f, g \rangle}{\|g\|_2^2}, f \right\rangle &= \langle f, f \rangle - \frac{\langle f, g \rangle}{\|g\|_2^2} \langle g, f \rangle \\
&= \|f\|_2^2 - \frac{|\langle f, g \rangle|^2}{\|g\|_2^2}
\end{aligned} \tag{4}$$

Combining (3) and (4) we get $0 \leq \|f\|_2^2 - |\langle f, g \rangle|^2 / \|g\|_2^2 \implies |\langle f, g \rangle|^2 \leq \|f\|_2^2 \|g\|_2^2 \implies |\langle f, g \rangle| \leq \|f\|_2 \|g\|_2$. $\square$

Proof of (c). Notice that

$$\begin{aligned}
\|f + g\|_2^2 &= \langle f + g, f + g \rangle \\
&= \langle f + g, f \rangle + \langle f + g, g \rangle \\
&= \langle f, f \rangle + \langle f, g \rangle + \langle g, f \rangle + \langle g, g \rangle \\
&= \|f\|_2^2 + \|g\|_2^2 + \langle f, g \rangle + \langle g, f \rangle \\
&\stackrel{\text{C.S.}}{\leq} \|f\|_2^2 + \|g\|_2^2 + 2\|f\|_2 \|g\|_2 \\
&= (\|f\|_2 + \|g\|_2)^2.
\end{aligned}$$

So we have $\|f + g\|_2^2 \leq (\|f\|_2 + \|g\|_2)^2 \implies \|f + g\|_2 \leq \|f\|_2 + \|g\|_2$. $\square$

Proof to (d). From (c) we have $\|f + g\|_2^2 = \|f\|_2^2 + \|g\|_2^2 + \langle f, g \rangle + \langle g, f \rangle$. By assumption we have $\langle f, g \rangle = 0 \implies \langle g, f \rangle = 0$. So we get $\|f + g\|_2^2 = \|f\|_2^2 + \|g\|_2^2$. $\square$

Proof of (e). We have $\|cf\|_2^2 = \langle cf, cf \rangle = c\bar{c}\langle f, f \rangle = |c|^2 \|f\|_2^2$. So we have $\|cf\|_2 = |c| \|f\|_2$. $\square$

Problem 2. Show that for any $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ we have $\|f\|_2 \leq \|f\|_\infty$.

Proof. It must be noted that for any $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ we have $\sup_{x \in \mathbb{R}} |f(x)|^2 = \sup_{x \in [0,1]} |f(x)|^2$ since f is $\mathbb{Z}$ -periodic. Notice $\|f\|_2^2 = \int_0^1 |f(x)|^2 dx \leq \int_0^1 \sup_{x \in [0,1]} |f(x)|^2 dx = \sup_{x \in [0,1]} |f(x)|^2$.

Lemma: If A is a set with real numbers greater than or equal to zero and is bounded, then $\alpha := \sup A$ exists and, in particular $\alpha^2 = \sup A^2$, where $A^2 := \{x^2 \mid x \in A\}$.

Proof. For all $x \in A$ $x \leq \alpha \implies x^2 \leq \alpha^2$. So α^2 is an upper bound for A^2 . Let β be another upper bound for A^2 . So for all $x \in A$, we have $x^2 \leq \beta$. So $x \leq \sqrt{\beta}$, i.e., $\sqrt{\beta}$ is an upper bound for A . So we must have $\sqrt{\beta} \geq \alpha \implies \beta \geq \alpha^2$. So $\alpha^2 = \sup A^2$. $\square$

By lemma above we have $\sup_{x \in [0,1]} |f(x)|^2 = \left(\sup_{x \in [0,1]} |f(x)| \right)^2$. So we get $\|f\|_2^2 \leq \left(\sup_{x \in [0,1]} |f(x)| \right)^2 = \|f\|_\infty^2$, i.e., $\|f\|_2 \leq \|f\|_\infty$. $\square$

Problem 3. Prove that L^2 metric d_{L^2} on $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ does indeed turn $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ into a metric space.

Proof. We need to show the metric space axioms are held.

- (1) We wish to show $d_{L^2}(f, g) > 0$ for $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ where f is not identically g . Indeed, since f not identically g , then there exists $x \in \mathbb{R}$ such that $|f(x) - g(x)| > 0 \implies |f(x) - g(x)|^2 > 0$. So $d_{L^2}(f, g) = \int_0^1 |f(x) - g(x)|^2 dx > 0$.
- (2) We wish to show for any $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, we have $d_{L^2}(f, f) = 0$. Take any $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. Then we have $|f(x) - f(x)|^2 = 0$ for all $x \in [0, 1]$. Hence $d_{L^2}(f, f) = \int_0^1 |f(x) - f(x)|^2 dx = 0$.
- (3) We wish to show for any $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ we have that $d_{L^2}(f, g) = d_{L^2}(g, f)$. Indeed for any $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ we have that $d_{L^2}(f, g) = \int_0^1 |f(x) - g(x)|^2 dx = \int_0^1 |g(x) - f(x)|^2 dx = d_{L^2}(g, f)$.
- (4) We wish to show that for any $f, g, h \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, $d(f, g) \leq d_{L^2}(f, h) + d_{L^2}(h, g)$. Take any $f, g, h \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$. By triangle inequality of the L^2 norm, we have, $d_{L^2}(f, g) = \|f - g\|_2 = \|(f - h) + (h - g)\|_2 \leq \|f - h\|_2 + \|h - g\|_2 = d_{L^2}(f, h) + d_{L^2}(h, g)$

Since all axioms are held, it follows that $(C(\mathbb{R}/\mathbb{Z}; \mathbb{C}), d_{L^2})$ is a metric space. $\square$

Problem 4. For any $n, m \in \mathbb{Z}$ we have $\langle e_n, e_m \rangle = 1$ when $n = m$ and $\langle e_n, e_m \rangle = 0$ when $n \neq m$. Also $\|e_n\|_2 = 1$.

Proof. Let $m, n \in \mathbb{Z}$. Then

$$\langle e_n, e_m \rangle = \int_0^1 e_n \overline{e_m} = \int_0^1 e^{2\pi i n x} e^{-2\pi i m x} dx = \int_0^1 e^{2\pi i (n-m)x} dx.$$

Notice that if $n = m$, then we have

$$\int_0^1 e^{2\pi i (n-m)x} dx = \int_0^1 e^0 dx = 1.$$

Notice that if $n \neq m$, then we have

$$\int_0^1 e^{2\pi i (n-m)x} dx = \left[\frac{e^{2\pi i (n-m)x}}{2\pi i (n-m)} \right]_0^1 = \frac{1}{2\pi i (n-m)} (e^{2\pi i (n-m)} - 1).$$

Since $n - m \in \mathbb{Z}$ then we have

$$e^{2\pi i (n-m)} = \cos(2(n-m)\pi) + i \sin(2(n-m)\pi) = 1 + i \cdot 0 = 1.$$

So we have,

$$\frac{1}{2\pi i (n-m)} (e^{2\pi i (n-m)} - 1) = 0.$$

It follows that

$$\langle e_n, e_m \rangle = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

Also we have

$$\|e_n\| = \sqrt{\langle e_n, e_n \rangle} = \sqrt{1} = 1.$$

□